

Introduction

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Sepsis is one of the most severe syndromes resulting in death, which is caused by a dysregulated host immune response to infection and leads to organ failure. Although a lot of research has been done in this area, early and precise diagnosis is still a very important, difficult task in sepsis treatment. Present clinical practice largely relies on non-specific biomarkers of inflammation (e.g., C-reactive protein [CRP], procalcitonin [PCT]) or slow blood culture-based methods.[1] These methods are non-specific for pathogens, they often cause a delay in targeted therapy, and they cannot differentiate bacterial sepsis from viral or fungal infections. The consequence is a heavy dependence on broad-spectrum antibiotics, which in turn leads to the problem of antimicrobial resistance (AMR).[2]

In the case of efficient point-of-care diagnostics and precision therapeutics, it is a very pressing issue to come up with pathogen-specific biomarkers that are indicative of the presence of bacteria in the blood and not the host responses.

Limitations of Current Diagnostic Approaches

Blood Cultures: The standard clinical method, but the sensitivity is quite low (30–50%), and it takes 24–72 hours for the results. The accuracy is also considerably lowered if antibiotics have been administered before the test.[3]

Molecular PCR-based Assays: These have higher sensitivity; however, they also require a pre-lysis step, high-quality nucleic acids, and a well-equipped laboratory. Besides, the specificity of primers and the inability to detect unknown or mutated strains limit them.[4]

Host-Derived Biomarkers (CRP, PCT, presepsin): These are pain sensors that only indicate the presence of inflammation but do not provide any clues about the causative agent. Besides, they can hardly distinguish bacterial from viral or fungal sepsis, limiting the specificity of therapy.[5]

Mass Spectrometry and ELISA Panels: Extensively processed samples are required; They are expensive, and they cannot be used for emergency situations as they do not meet the rapid turnaround requirements.[6]

Therefore, current sepsis diagnostics tools are either too slow, too nonspecific, or too resource-intensive to be really helpful as frontline tools.

Rationale for Bacterial-Specific Biomarkers

Bacteria differ from viruses and fungi in that they have unique surface-exposed molecules that can be direct and discriminative sepsis biomarkers.[7] Besides being structurally conserved, immunologically significant, and easily accessible for detection in blood samples by techniques such as surface-enhanced Raman spectroscopy (SERS), immunoassays, or biosensors, these markers also possess such features. By adopting bacterial outer membrane proteins, lipopolysaccharide (LPS) fragments, or capsular antigens as diagnostic targets, they can move from detecting indirect inflammation to direct bacterial fingerprinting.[6]

Potential Bacterial Surface Biomarkers in Sepsis

1. Lipopolysaccharides (LPS) – Gram-Negative Bacteria. Endotoxins that are released into the blood circulation during bacteremia. The lipid A and O-antigen peculiarities can be the base for separation of different Gram-negative species. The indication of septic shock and the activation of the host immune system can be very strong.[8]
2. Lipoteichoic Acid (LTA) – Gram-Positive Bacteria. LTA is a component structurally different from LPS that is attached to the cell wall and unique to the Gram-positive pathogens. It becomes free in blood during infection and causes TLR-mediated inflammation.[9]
3. Outer Membrane Proteins (OMPs), Examples: OmpA, OmpC, OmpF (Enterobacteriaceae), porins in *Pseudomonas aeruginosa*. Highly immunogenic, that is, they are surface-exposed and species-specific. It is possible for them to be not only diagnostic antigens but also therapeutic targets at the same time.[10]
4. Capsular Polysaccharides (CPS), Detected in *Klebsiella pneumoniae*, *Streptococcus pneumoniae*, and *Neisseria meningitidis*. Their epitope patterns are serotype-specific allowing differentiation of pathogens. During infection, they become shed into the blood and thus can be detected by the use of antigen capture methods.[11]
5. Peptidoglycan Fragments (PGN, Muramyl Dipeptide), New fragments are generated during the bacterial growth and lysis. Not present in viral/fungal infections, hence they are highly specific to bacterial sepsis.[12]
6. Flagellin and Pili Proteins, Most that may be found in the cells of bacteria that can move are structural proteins uniquely. The host innate immune system (TLR5, mannose receptors) recognizes them. They can act as biomarkers indicating that the bacteria are present in real time.[13]
7. Exotoxins/Secreted Proteins (e.g., α -hemolysin, pneumolysin), Pathogen-specific proteins that can be directly detected in plasma. They are signs of an active infection rather than residual inflammation.[14]

Conclusion

Instead of relying on late, host-derived inflammation markers, direct detection of bacterial fingerprints can enable rapid identification and differentiation of Gram-positive versus Gram-negative infections. Techniques such as Surface-Enhanced Raman Spectroscopy (SERS) and other nano-enabled biosensors are uniquely suited for this task, as they can capture molecular vibrational signatures or antigen-specific interactions within minutes. Our primary focus is on SERS-based detection, given its ability to provide rapid, label-free, and highly sensitive molecular fingerprinting.[15,16]

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